A Short Primer of Economics Experiments

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May 29, 2026
2026 정책실험포럼

Economics experiments

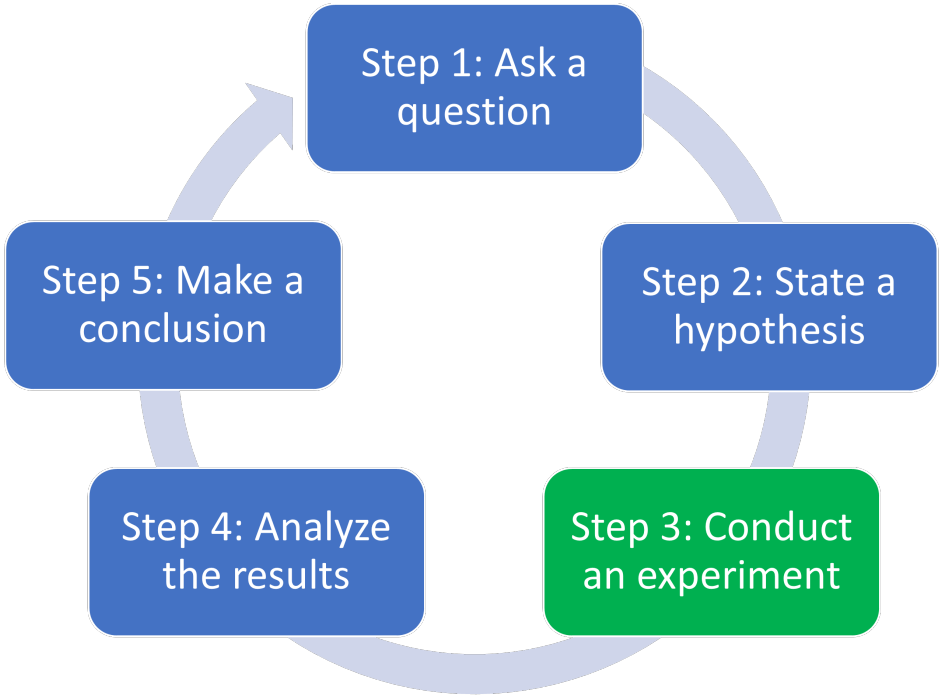
Behavioral economics $\neq$ experimental economics

- Experimental economics is a small field of economics that considers how experiments are properly designed and analyzed.
- An experiment is a method to collect data in a controlled environment. Experiments are used by many behavioral economists, but they use it as a method, not a field of study.

  (Think about econometrics. It is a research field as well as a method for researchers in other fields.)

Economics experiments are

- never like YouTubers' "social experiments" (or pranks);
- less like survey-based psychology experiments;
- more like **science experiments**.



A typical controlled experiment

Suppose you want to test a new drug. How would you do?

- Recruit human subjects.
- Randomly split them into two groups.
- One group, called the treatment group, gets the drug. The other group, called the control group, gets a placebo.
- Compare the observations of interest between two groups.

What's the difference? The observations of interest may be subject to the existence (and behaviors) of the experimenter, and the subject's will *can* alter the observations. (Overall it is called the **experimenter demand effect**. I will discuss it later.)

Types of experiments

Depending on the trade-off between *control* and *realism*, economics experiments take different forms.

- **Lab experiments**: Subjects (typically students) make decisions in a tightly controlled environment. Maximum control; external validity is the usual concern.
- **Lab-in-the-field**: The same laboratory protocols, but run with the population of interest (e.g., CEOs, traders, procurement officers, or compliance managers). Preserves experimental control while making the subject pool more relevant to the policy question.
- **Randomized controlled trials (RCTs)**: Random assignment of actual policies or interventions to real agents in the field. Lower control, but directly estimates policy impact at scale.

These are complementary, not ranked; the right venue depends on the research question.

Distinctive traditions of experimental economics

And recent discussions

- No deception (Most distinctive)
- Incentives
- Abstract framing
- Control of beliefs
- Repetition or stationary replication
- Complete reporting of methods
- WEIRD sample (Least distinctive)

No deception (1/2)

Economists never deceive experimental participants.

- "Deception" defined as outright lying; incomplete information not considered deception.
- Usually justified as maintaining reputation for no deception, for experimental control of subjects' beliefs.
- Externality from deception for other experiments can be very harmful.

No deception (2/2)

Some social psychologists deceive subjects as a matter of course. (YouTubers' "social experiments" use deception explicitly.)

- There are a number of (psychology) experiments on the effects of deception.
- Some find that subjects under-report levels of suspicion in "suspicion probes," and become more suspicious in future experiments.
- Evidence of reputation effects

Notes: People have substantially heterogeneous standards in "gray areas" (Charness and Samek, 2022). Are omitting information, misleading without an explicit lie, or using confederates okay? What about fooling people outside of the lab (Bertrand and Mullainathan, 2004; Bursztyn et al, 2021)?

Incentives (1/2)

All (lab) experimental studies published in the AER during 1970–2015 used financial incentives.

- Partly for reasons of making "economic situations".
- Partly for experimental control of subjects' motivations.
- Many psychologists offer course credit instead.
- Camerer and Hogarth (1999) reviewed 74 studies that use high, low, and no incentives, and reported (i) incentives affect performance in tasks where effort helps performance, e.g., judgment, memory, and clerical tasks, (ii) incentives do not affect performance in tasks where intrinsic motivation is high, or effort isn't helpful; (iii) high incentives can hurt performance in some tasks, e.g., choking under pressure, self-conscious anxiety about performance.

Incentives (2/2)

In general, higher incentives:

- Reduce variance in behavior (i.e., less noise / confusion).
- Reduce the influence of self-presentation concerns, e.g., people say they are more generous and risk-taking than they are.
- Otherwise, have little effect on average behavior.

Note: There are new pieces of evidence that the incentives may not matter that much (Rubinstein, 2012; Voslinsky and Azar, 2021). In general, incentives do not worsen the quality of the observations, but sometimes it does (Danz, Vesterlund, and Wilson, 2022).

Abstract framing (1/2)

Most economics experiments attempt to have "no context," e.g., Person A and Person B instead of Police and Thief.

- Partly because economic theory being tested has "no context." Stag hunt and bank run are different contexts, but the same model can capture the common essence.
- Partly because according to game theory, labels may serve as a coordinating device.

  (Imagine I label one action as 'clever' and another as 'stupid.')

Subjects will anyway understand the experiment in their own contextual view. Still, abstract framing needs to be maintained to increase validity of the controlled environment.

Abstract framing (2/2)

Some recent experimental studies use more specific framing.

- Sometimes abstract framing decreases the experiment participants' problem-solving ability.
- Sometimes research question might be tightly related to a specific situation.
- We still need to ensure for the framing not to be experimenter-demand intended.

Vignette studies, as the name tells, adopt very specific framing. They are useful to deliver clear messages, but some claim they are nothing but thought experiments.

Commonly raised (but answered) questions

- Do subjects really "think" to make decisions and "work hard" to earn in the lab? At the worst, they could buy a pizza slice, and at best they could treat themselves nice dinner...
- Aren't the stakes too small? Will I be risk averse when I bet 30 cents to get 60 cents with some probability?

⇒ Early experimental economists replicated existing experiments at the countries where the price is low (e.g., $20 could be the country's average monthly income.)

⇒ Vesterlund et al. (2016) report inelastic effort provision with respect to incentives.

Commonly raised (but partly answered) questions

- In real life situations, there is no such "experimenter" who will analyze my choices. What if I would change my decisions due to the existence of the experimenter? What if the experimenter implicitly "demands" me doing something?

$\Rightarrow$ It is called the experimenter demand effect. Economists have deliberately designed experiments to minimize such unwanted effects (e.g., double-blind system in ultimatum game).

$\Rightarrow$ The treatment effects are still valid if the **demand effects are orthogonal** to the treatment conditions.

Consider this: 'White coat hypertension' exists, but it is orthogonal to the effectiveness of a new drug.

$\Rightarrow$ However, **if the demand effects differ across treatment conditions**, it may be a severe threat to internal validity.

Consider this: In an information-provision experiment, information may implicitly guide what recipients "should" do.

Encouragement: Conduct experiments

When theory is limited to tell what's going on;

- When theory predicts that everything between A and B is possible?
- Among two theories for the same observation, what's the main driver?

When you cannot do comparative statics with empirical data;

- When the FOMC has more members? The effect when we change the voting rule for international climate agreements? The impact of a mandatory social mix?

When evidence for policy suggestions is insufficient;

- How should leniency rewards be structured to break cartels? Which disclosure format reduces consumer harm from drip pricing? Does banning specific dark patterns actually improve welfare, or do firms substitute to equivalently harmful designs?